Rajat Ghosh, Ph.D.

+1 (404) 697-5789 · rajat.ghosh11@gmail.com · LinkedIn · Google Scholar · Website

Professional Summary

Senior technical leader building company-scale GenAI platforms for post-training and evaluation from research to production. Focus on strategic value creation from GenAI using experimentation. Set long-term technical vision in high-ambiguity frontier domains (agentic systems, RAG, developer productivity, AI safety), while scaling high-performing teams and partnering with Product, Infra, Legal, and executive leadership.

Professional Experience

Technical Director, AI — Nutanix, *September 2025 - Present*

As a Technical Director, I lead the post-training and evaluation platform.

- Hired and managed a team of 7 early-career engineers to build a sovereign AI platform.
- Architected post-training platform (GRPO and SFT) on hybrid multi-cloud environments.
- Led architecture, evaluation, and production rollout for coding agents.
- Architected evaluation and multimodal data pipelines at MLCommons.

Staff Data Scientist — Nutanix, *May 2021 - August 2025*

As Tech Lead Data Scientist, I spearheaded the design and delivery of two production-ready GenAI agents for customer service and developer productivity.

- Architected internal coding assistant agents utilizing open-source LLMs and advanced agentic patterns such as RAG, tool-calling, Reflexion, and adoption of third party AI coding assistants ($100M annual impact from 20% improvement in developer productivity).
- Developed customer support assistants using open source LLMs and retrieval augmented generation (RAG) with open-source VectorDB ($20M annual impact from from support engineering productivity).
- Engineered reinforcement learning strategies for distributed infrastructure management including Kubernetes, Cassandra, storage tiering, data warehousing, and load balancing.

Co-Founder & CEO — AdeptDC, Inc. *July 2015 – April 2021*

I founded a company to commercialize PhD research in reinforcement learning and data center energy efficiency, with a core innovation in closed-loop control of data center cooling systems using chip-level temperature signals from physical sensors.

- Built and led a 7-member team delivering AI-driven optimization solutions to enterprises.
- Secured $1M in venture and NSF funding; exited via acquisition.

Education

- **Ph.D.**, Georgia Institute of Technology, Atlanta, 2013
- **M.S.**, Georgia Institute of Technology, Atlanta, 2010
- **B.S.**, Indian Institute of Technology, Kharagpur, 2008

Honors and Awards

- ICML Silver Reviewer, 2026
- NSF Small Business Innovation Research Award, 2015
- NSF I-Corps Award, 2014

Patents

- **R. Ghosh** and Y. K. Joshi, *Systems and Methods for Intelligent Controls for Optimal Resource Allocation for Data Center Operations*, US Patent 10,439,912, Oct. 2019. (Link)

Selected Publications — LLMs & Alignment

- P. Shah, **R. Ghosh**, A. Singhal, and D. Dutta, *RANGER: Repository-level Agent for Graph-Enhanced Retrieval*, AgenticSE Workshop, KDD 2026.
- **R. Ghosh** and D. Dutta, *Action Shapley: A training data selection metric for Training World Models for Reinforcement Learning*, Workshop on World Models: Understanding, Modelling and Scaling, ICLR 2026.
- **R. Ghosh**, V. Bhargava, and D. Dutta, *Compute-Efficient GRPO Training*, Workshop on Scaling Post-training for LLM, ICLR 2026.
- K. Pereira, N. Sinha, **R. Ghosh**, and D. Dutta, *CR-Bench: Evaluating the Real-World Utility of AI Code Review Agents*, Workshop on Agents in the Wild: Safety, Security, and Beyond, ICLR 2026.
- A. Lalan, **R. Ghosh**, and D. Dutta, *UT-Evolve: An Evolutionary Agent for Unit Test Writing*, Workshop on Multi-Agent Learning and Its Opportunities in the Era of Generative AI, ICLR 2026.
- A. Singhal, **R. Ghosh**, et al., *Code2JSON: Can a Zero-Shot LLM Extract Code Features for Code RAG?*, Workshop on Deep Learning for Code, ICLR 2025.
- V. Bhargava, **R. Ghosh**, and D. Dutta, *CPP-UT-Bench: Can LLMs Write Complex Unit Tests in C++?*, Workshop on Statistics in LLMs, NeurIPS 2024.
- A. Khera, **R. Ghosh**, and D. Dutta, *Efficient Alignment of Large Language Models via Data Sampling*, ENLSP Workshop (PMLR), NeurIPS 2024.

Consortium Publications — MLCommons

- Contributor (1 of 42 authors), *AILuminate Security: Jailbreak Benchmark v0.5*, MLCommons, 2025.
- Contributor (1 of 82 authors), *AILuminate: AI Risk and Reliability Benchmark v1.0*, MLCommons, 2025.
- Contributor (1 of 100 authors), *AI Safety Benchmark v0.5*, MLCommons, 2024.

Selected Publications — Doctoral Research (Foundations)

- **R. Ghosh** and Y. Joshi, *Proper Orthogonal Decomposition-Based Modeling Framework for Improving Spatial Resolution of Measured Temperature Data*, IEEE Transactions on Components, Packaging and Manufacturing Technology, 2014.
- **R. Ghosh** and Y. Joshi, *Rapid Temperature Predictions in Data Centers Using Multi-Parameter Proper Orthogonal Decomposition*, Numerical Heat Transfer, Part A: Applications, 2014.
- **R. Ghosh** and Y. Joshi, *Error Estimation in POD-Based Dynamic Reduced-Order Thermal Modeling of Data Centers*, International Journal of Heat and Mass Transfer, 2013.
- **R. Ghosh**, G. A. Buxton, O. B. Usta, A. C. Balazs, and A. Alexeev, *Designing Oscillating Cilia That Capture or Release Microscopic Particles*, Langmuir, 2010.

Invited Talks

- Agentic AI: Future Challenges and Opportunities, AAAI Workshop, March 2025. (Link)

Review & Service

Reviewer for ICML, SIGIR, ICLR (2026), NeurIPS (2025), AAAI (2025), IEEE CPMT Journal.